

Bayesian inference in non-Gaussian time series with global-local shrinkage process priors

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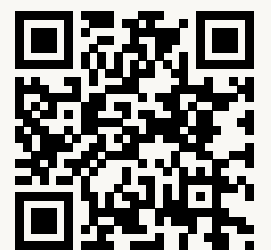
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Slides



 **Julia packages (unoptimized, ongoing)**



Outline

0. Time-varying parameters and state-space models
1. Dynamic shrinkage process priors
2. Time-varying multi-seasonal ARMA
3. Non-Gaussian models
4. Posterior inference

Time-varying parameter models

- Regression with **time-varying parameters**

$$\begin{aligned} y_t &= \mathbf{x}_t^\top \boldsymbol{\beta}_t + \varepsilon_t & \varepsilon_t &\stackrel{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2) \\ \boldsymbol{\beta}_t &= \boldsymbol{\beta}_{t-1} + \boldsymbol{\nu}_t & \boldsymbol{\nu}_t &\stackrel{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_\nu) \end{aligned}$$

- **Linear Gaussian state-space model**

Observation model: $\mathbf{y}_t = \mathbf{C}_t \boldsymbol{\theta}_t + \boldsymbol{\varepsilon}_t \quad \boldsymbol{\varepsilon}_t \stackrel{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon,t})$

Transition model: $\boldsymbol{\theta}_t = \mathbf{A}_t \boldsymbol{\theta}_{t-1} + \boldsymbol{\nu}_t \quad \boldsymbol{\nu}_t \stackrel{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\nu,t})$

with state $\boldsymbol{\theta}_t = \boldsymbol{\beta}_t$, $\mathbf{C}_t = \mathbf{x}_t^\top$ and $\mathbf{A}_t = \mathbf{I}_p$.

Regression with time-varying parameters

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Taxonomy of state-space models

- **Linear Gaussian**

$$\begin{aligned} \mathbf{y}_t &= \mathbf{C}_t \boldsymbol{\theta}_t + \boldsymbol{\varepsilon}_t & \boldsymbol{\varepsilon}_t &\stackrel{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon,t}) \\ \boldsymbol{\theta}_t &= \mathbf{A}_t \boldsymbol{\theta}_{t-1} + \boldsymbol{\nu}_t & \boldsymbol{\nu}_t &\stackrel{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\nu,t}) \end{aligned}$$

- **Non-linear Gaussian** (additive) models

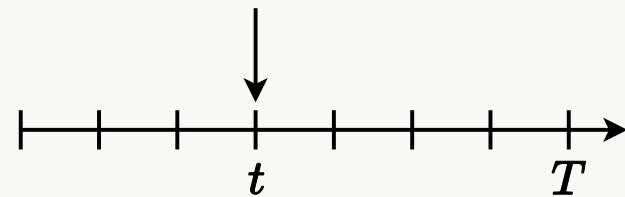
$$\begin{aligned} \mathbf{y}_t &= \mathbf{c}_t(\boldsymbol{\theta}_t) + \boldsymbol{\varepsilon}_t & \boldsymbol{\varepsilon}_t &\stackrel{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon,t}) \\ \boldsymbol{\theta}_t &= \mathbf{a}_t(\boldsymbol{\theta}_{t-1}) + \boldsymbol{\nu}_t & \boldsymbol{\nu}_t &\stackrel{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\nu,t}) \end{aligned}$$

- **Non-Gaussian** - general distributions

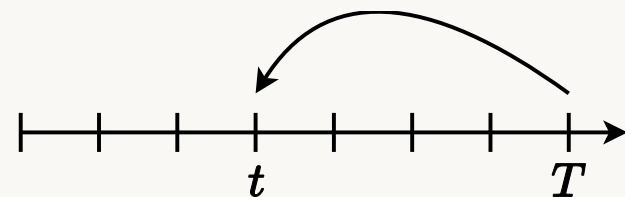
$$\begin{aligned} \mathbf{y}_t &\sim p_t(\mathbf{y}_t | \boldsymbol{\theta}_t) \\ \boldsymbol{\theta}_t &\sim q_t(\boldsymbol{\theta}_t | \boldsymbol{\theta}_{t-1}) \end{aligned}$$

Bayesian inference in state-space models

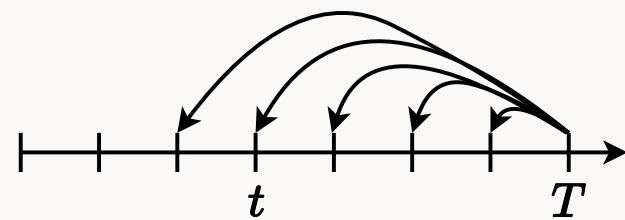
- **Filtering posterior** (instantaneous) $p(\boldsymbol{\theta}_t | y_{1:t})$



- **Smoothing posterior** (retrospective) $p(\boldsymbol{\theta}_t | y_{1:T})$



- **Joint smoothing posterior** $p(\boldsymbol{\theta}_{1:T} | y_{1:T})$



Global-local shrinkage priors - regression [1, 2]

- Linear Gaussian regression with **horseshoe prior**

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2)$$

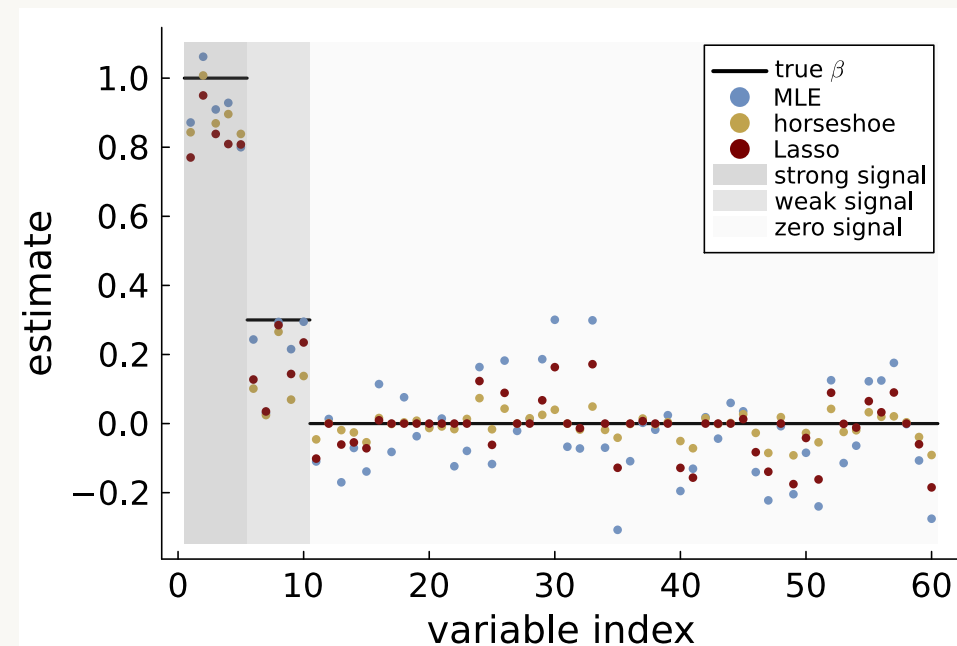
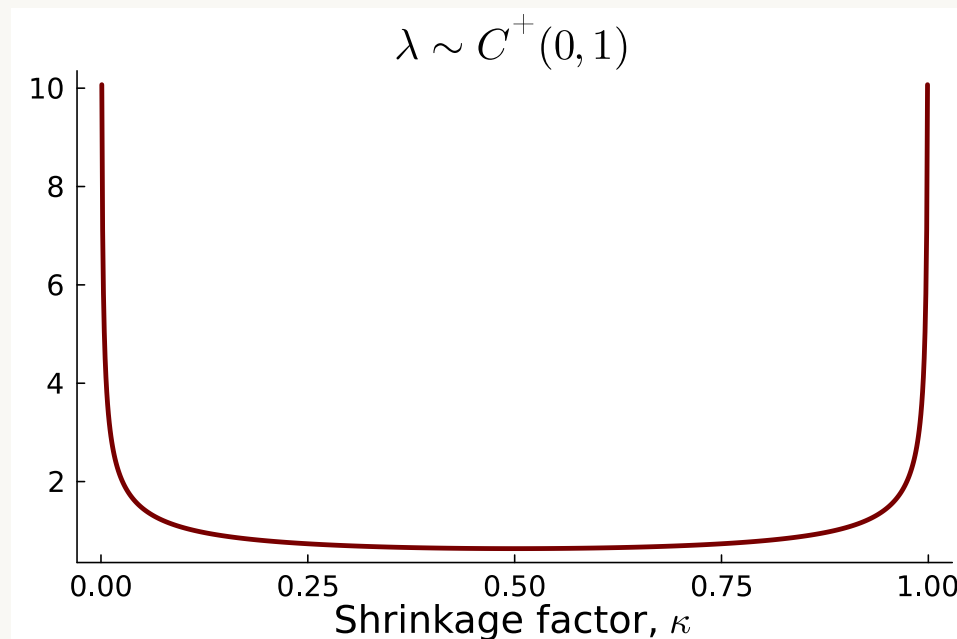
$$\beta_j \stackrel{\text{iid}}{\sim} N(0, \tau^2 \lambda_j^2)$$

$$\tau \sim C^+(0, 1)$$

$$\lambda_j \stackrel{\text{iid}}{\sim} C^+(0, 1)$$

- Posterior mean (for orthogonal $\mathbf{x}$) equals shrunk ML estimator

$$\tilde{\beta}_j = (1 - \kappa_j) \hat{\beta}_j$$



Dynamic global-local shrinkage priors for time series

- **Local level model - dynamic shrinkage process (DSP) prior**

$$\begin{aligned}
 y_t &= \mu_t + \varepsilon_t & \varepsilon_t &\stackrel{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2) \\
 \mu_t &= \mu_{t-1} + \nu_t, & \nu_t &\stackrel{\text{ind}}{\sim} N(0, \tau^2 \exp(h_t)) \\
 h_t &= \rho h_{t-1} + \eta_t, & \eta_t &\stackrel{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1) \\
 \tau &\sim C^+(0, 1)
 \end{aligned}$$

- **Global variance τ^2**
- **Local variance $\lambda_t^2 = \exp(h_t)$**
- DSP becomes horseshoe when $\rho = 0$:

$$h_t \sim Z(\alpha, \beta, 0, 1) \iff \lambda_t = \exp(h_t/2) \sim C^+(0, 1)$$

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Logistic-Beta distribution (Fisher's Z)

- $X \sim \text{Beta}(\alpha, \beta)$ then

$$Z := \log \left(\frac{X}{1-X} \right) \sim Z(\alpha, \beta, 0, 1)$$

- Location-scale:

$$\mu + \sigma Z \sim Z(\alpha, \beta, \mu, \sigma)$$

- **Heavy tails** - log density decays linearly.

- Mean-variance mixture of Gaussians [3]

$$\begin{aligned} Z|\xi &\sim N(\xi^{-1}(\alpha - \beta)/2, \xi^{-1}) \\ \xi &\sim \text{PolyaGamma}(\alpha + \beta, 1) \end{aligned}$$

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Multi-seasonal ARMA with DSP prior [4]

- Seasonal AR with DSP evolution

$$\phi_{p,t}(L)\Phi_{P,t}(L^s)(y_t - \mu_t) = \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} N(0, \sigma_t^2)$$

- SAR(1,1) as a **nonlinear Gaussian regression**

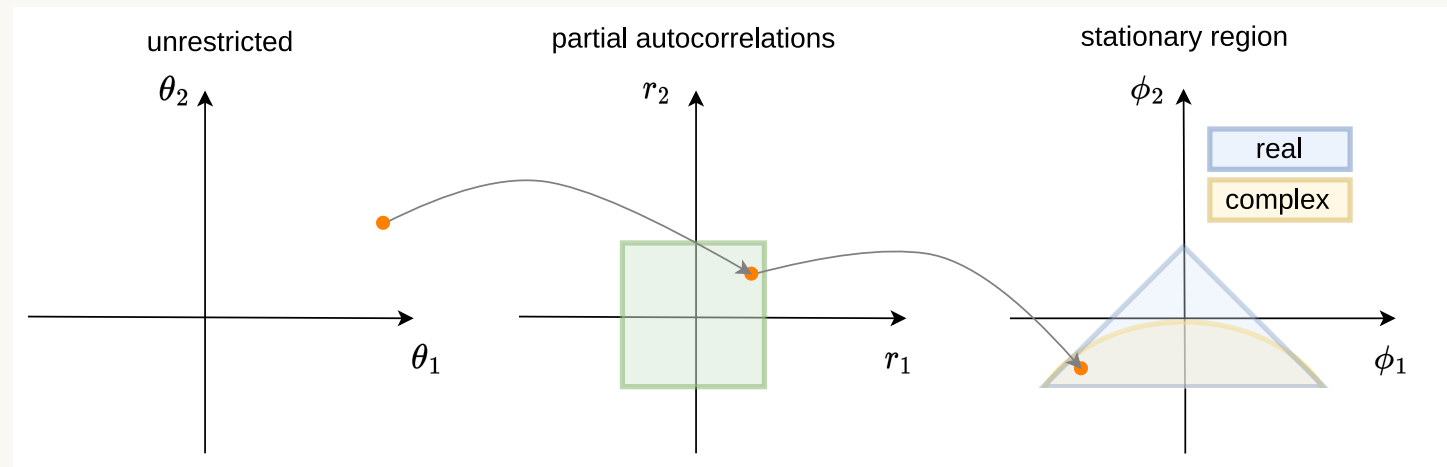
$$y_t = \mathbf{z}_t^\top \tilde{\boldsymbol{\phi}} + \varepsilon_t$$

$$\tilde{\boldsymbol{\phi}} = (\phi_1, \Phi_1, -\phi_1 \Phi_1)^\top \text{ and } \mathbf{z}_t = (x_{t-1}, x_{t-s}, x_{t-(1+s)})^\top$$

- Extensions
 - ✓ **Multi-seasonal** (daily, weekly, yearly cycle)
 - ✓ ARMA and Exact likelihood
 - ✓ Stochastic volatility with DSP priors
 - Multi-seasonal forecasting
 - Multi-seasonal VAR

Stability restrictions

Transformation to stability at every t



Lemma - uniform prior over the stability region [4, 5]

The AR parameters ϕ are uniformly distributed over the stability region if, independently,

$$\theta_k \sim \begin{cases} t(k+1, 0, \frac{1}{\sqrt{k+1}}) & \text{for } k \text{ is odd} \\ t_{\text{skew}}(\frac{k}{2}, \frac{k+2}{2}, 0, \frac{1}{\sqrt{k+1}}) & \text{for } k \text{ is even} \end{cases}$$

- Transformation to partial autocorr matters

- **Monahan**

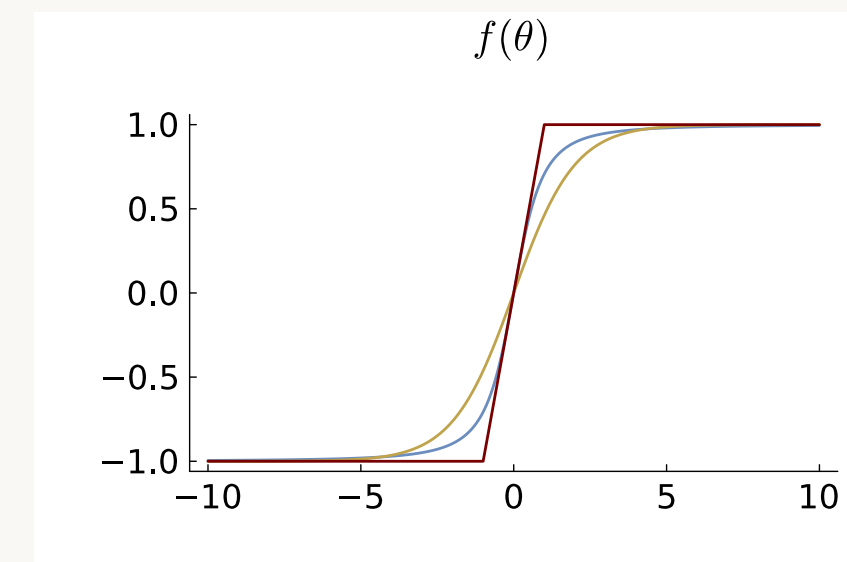
$$r = \frac{\theta}{\sqrt{1 + \theta^2}}$$

- **Tanh**

$$r = \frac{e^{\theta} - e^{-\theta}}{e^{\theta} + e^{-\theta}}$$

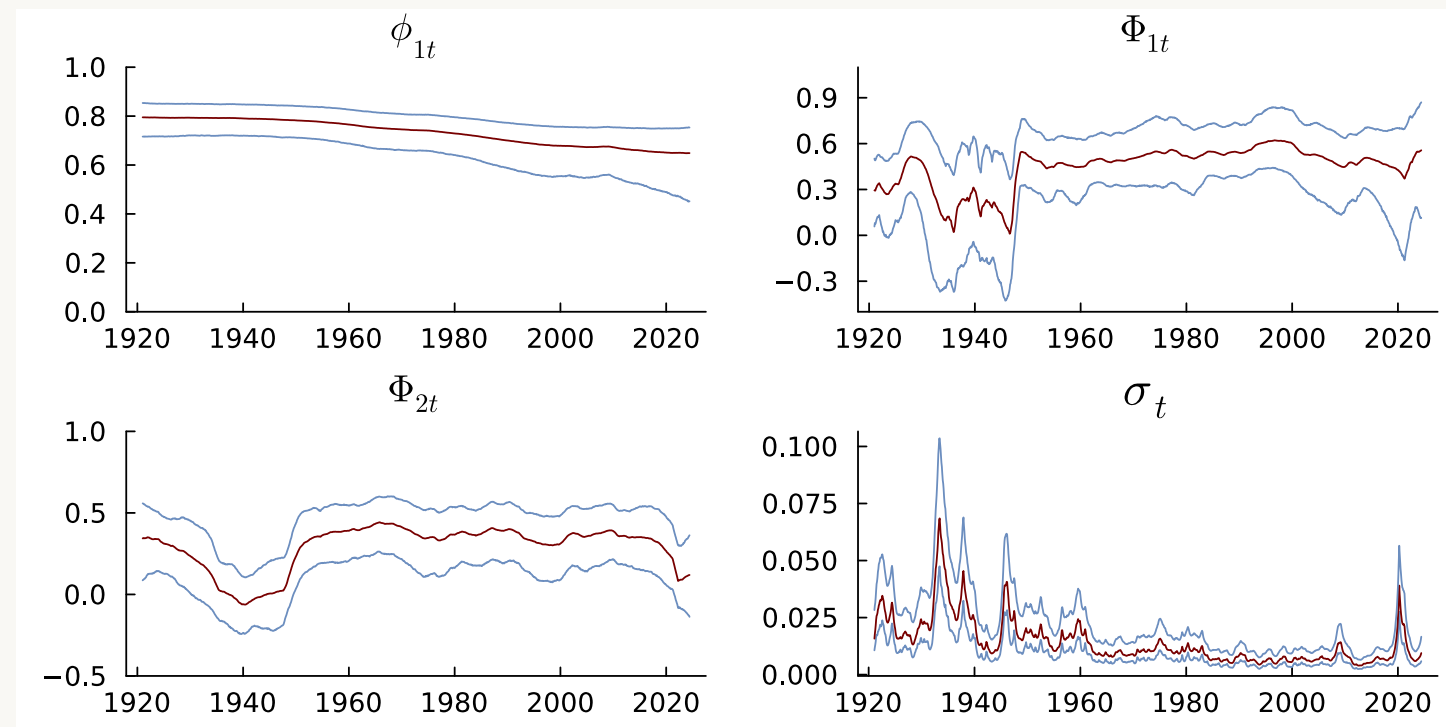
- **Hard-tanh**

$$r = \begin{cases} -(1 - \epsilon) & \text{if } \theta \leq -(1 - \epsilon) \\ \theta & \text{if } -(1 - \epsilon) < \theta < 1 - \epsilon \\ 1 - \epsilon & \text{if } \theta \geq 1 - \epsilon \end{cases}$$

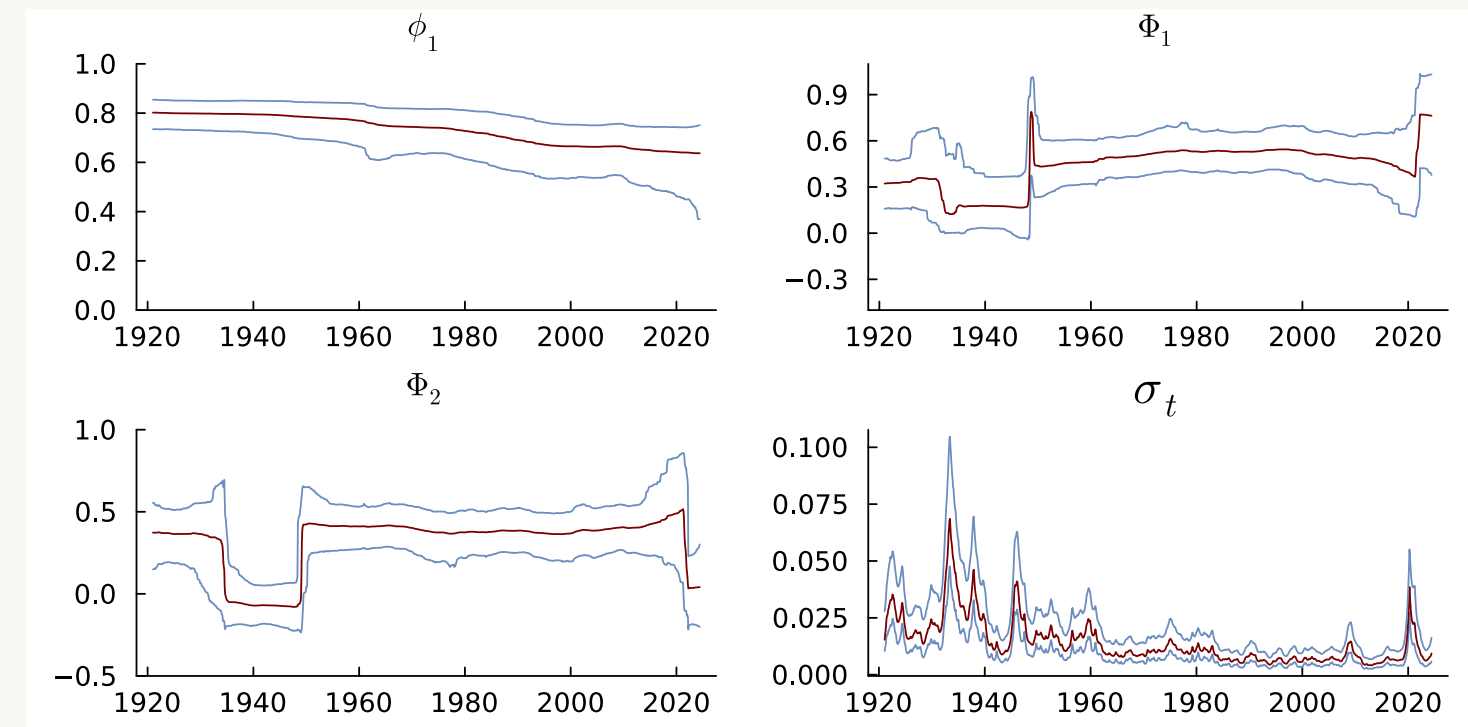


Seasonal AR with DSP prior - US industrial production 1919-2024

SAR(1,2) - Homoscedastic Gaussian



SAR(1,2) - DSP



Time-varying generalized linear models with DSP evolution

- **GLMs** with time-varying parameters

$$y_t | \mathbf{x}_t \sim \text{ExpFamily} \left(k^{-1}(\mathbf{x}_t^\top \boldsymbol{\beta}_t), \varphi_t \right)$$

$$\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} + \boldsymbol{\nu}_t, \quad \boldsymbol{\nu}_t \stackrel{\text{iid}}{\sim} N(\mathbf{0}, \text{Diag}(\exp(\boldsymbol{\tau}^2 \odot \mathbf{h}_t)))$$

$$\mathbf{h}_t = \Phi \mathbf{h}_{t-1} + \boldsymbol{\eta}_t, \quad \eta_{tk} \stackrel{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1)$$

- **Poisson regression** with DSP prior

$$y_t | \mathbf{x}_t \sim \text{Pois}(\mu_t = \exp(\mathbf{x}_t^\top \boldsymbol{\beta}_t))$$

- **Beta regression** with DSP prior

$$y_t | \mathbf{x}_t, z_t \sim \text{Beta}(\mu_t = \text{logistic}(\mathbf{x}_t^\top \boldsymbol{\beta}_t), \psi_t = \exp(\mathbf{z}_t^\top \boldsymbol{\gamma}_t))$$

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Bayesian inference in state-space models with DSP prior

- Joint smoothing posterior ($\boldsymbol{\psi}$ holds all static parameters):

$$p(\boldsymbol{\theta}_{1:T}, \boldsymbol{h}_{1:T}, \boldsymbol{\psi} | \boldsymbol{y}_{1:T})$$

- Gibbs sampling/MCMC
- Particle MCMC
- Hamiltonian Monte Carlo (HMC)
- Variational approximations
- Integrated Nested Laplace Approximation (INLA)?

Gibbs sampling in linear Gaussian model with DSP prior

- **Polya-Gamma augmentation** to turn $\mathbf{Z}$ -distribution into conditionally Gaussian. [6, 7]
- **Square-and-log trick** to turn variance $\exp(h_t)$ into a mean h_t . [8]
- Fast sampling of $\mathbf{h}_{1:T}$ using **sparse multivariate Gaussian** [9].
- **Forward-filtering-backward sampling** (FFBS) to sample $\boldsymbol{\theta}_{1:T}$ [10, 11]

Gaussian approximation filters

- **Forward-filtering-backward sampling** for linear Gaussian models [10, 11]
 - Run Kalman filter $p(\boldsymbol{\theta}_t | \mathbf{y}_{1:t})$ forward for $t = 1, 2, \dots, T$
 - Sample $p(\boldsymbol{\theta}_t | \boldsymbol{\theta}_{t+1}, \mathbf{y}_{1:T})$ backward for $t = T, T-1, \dots, 1$
- Non-linear/non-Gaussian: Kalman filter and FFBS not applicable.
- General idea: replace the filtering step with an approximate filter.
- **Prior propagation**: approximate the joint distribution

$$\begin{pmatrix} \boldsymbol{\theta}_t \\ \boldsymbol{\theta}_{t-1} \end{pmatrix} | \mathbf{y}_{1:t-1} \stackrel{\text{approx}}{\sim} N \left[\begin{pmatrix} \mathbb{E}(\boldsymbol{\theta}_t) \\ \mathbb{E}(\boldsymbol{\theta}_{t-1}) \end{pmatrix}, \begin{pmatrix} \mathbb{V}(\boldsymbol{\theta}_t) & \mathbb{C}(\boldsymbol{\theta}_t, \boldsymbol{\theta}_{t-1}) \\ \mathbb{C}(\boldsymbol{\theta}_t, \boldsymbol{\theta}_{t-1})^\top & \mathbb{V}(\boldsymbol{\theta}_{t-1}) \end{pmatrix} \right]$$

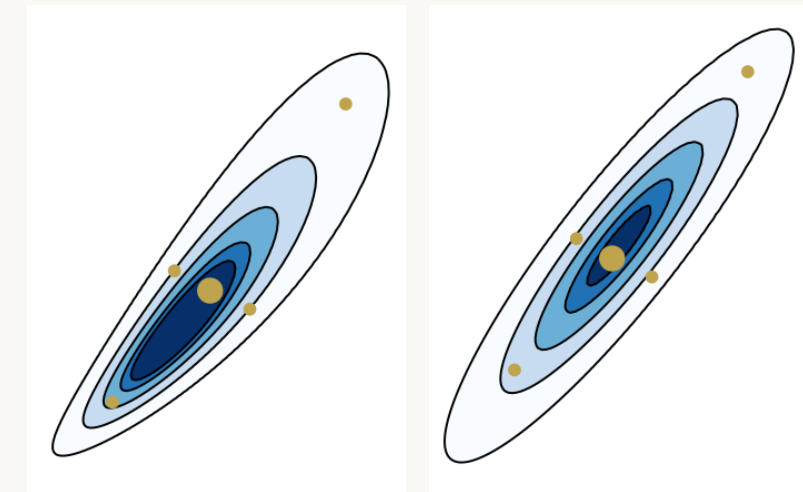
- **Observation update**: approximate the joint distribution

$$\begin{pmatrix} \mathbf{y}_t \\ \boldsymbol{\theta}_t \end{pmatrix} | \mathbf{y}_{1:t-1} \stackrel{\text{approx}}{\sim} N \left[\begin{pmatrix} \mathbb{E}(\mathbf{y}_t) \\ \mathbb{E}(\boldsymbol{\theta}_t) \end{pmatrix}, \begin{pmatrix} \mathbb{V}(\mathbf{y}_t) & \mathbb{C}(\mathbf{y}_t, \boldsymbol{\theta}_t) \\ \mathbb{C}(\mathbf{y}_t, \boldsymbol{\theta}_t)^\top & \mathbb{V}(\boldsymbol{\theta}_t) \end{pmatrix} \right]$$

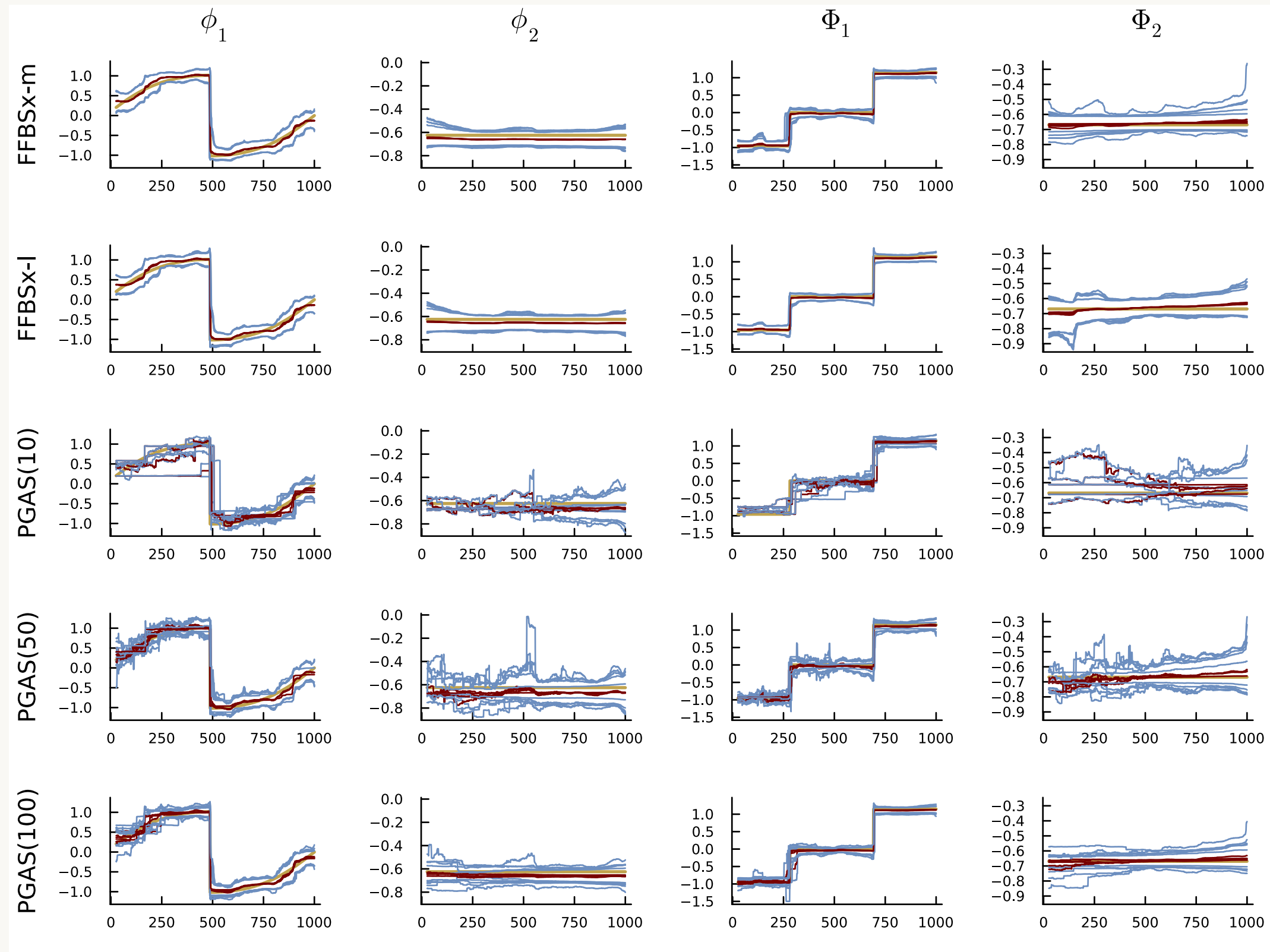


Gaussian approximation filters

- **Extended Kalman filter**
 - Linearize measurement/transition models
 - Apply standard Kalman filter
- **Unscented Kalman filter**
 - Propagate sigma point approx of prior/posterior
- **Prior linearization filter**
 - Linearize by statistical linear regression



Simulated example SAR(2,2) - MCMC convergence



Iterated Gaussian approximation filters

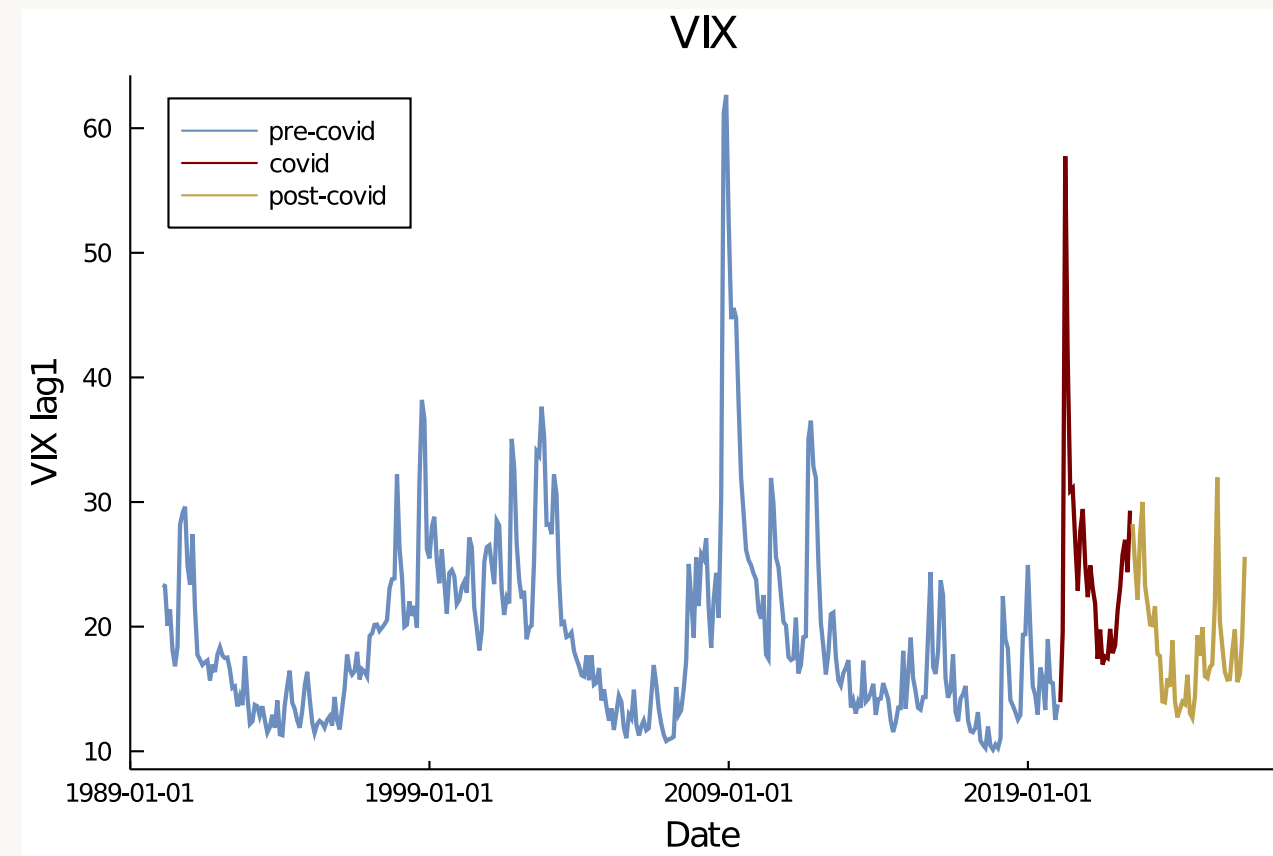
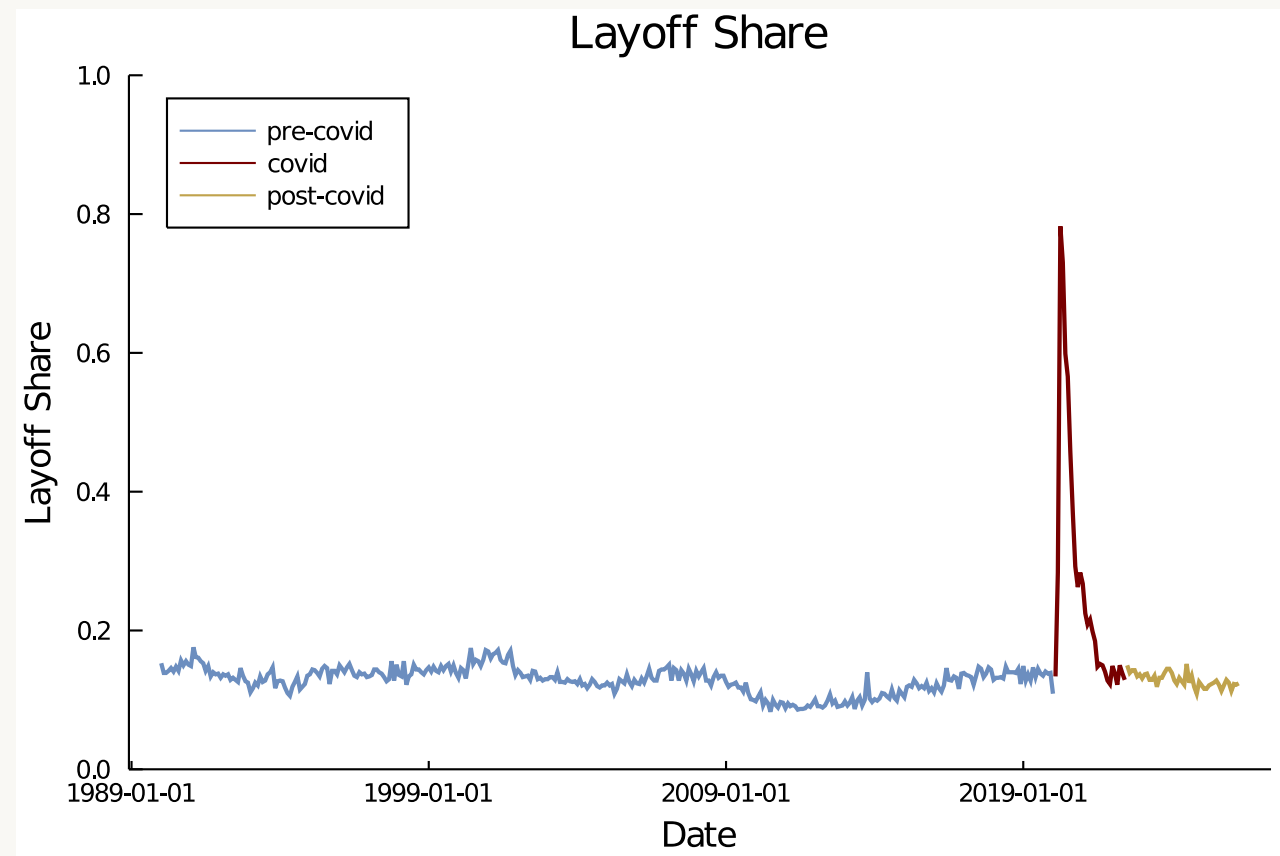
- Nonlinear Gaussian models
 - **Iterated extended Kalman filter** [12]
 - **Iterated unscented Kalman filter** [13]
 - **Posterior linearization filter** [14]
- Non-Gaussian models
 - **Posterior linearization filter - conditional moments version** [15]
 - Statistical linearization using $\mathbb{E}(Y|\mathbf{x})$ and $\mathbb{V}(Y|\mathbf{x})$.
 - **Laplace filter** [16, 17]
 - Direct Gaussian approximation of filtering posterior at every t

$$\boldsymbol{\theta}_t | \mathbf{y}_{1:t} \stackrel{\text{approx}}{\sim} N(\tilde{\boldsymbol{\theta}}_{t|t}, \tilde{\boldsymbol{\Sigma}}_{t|t})$$

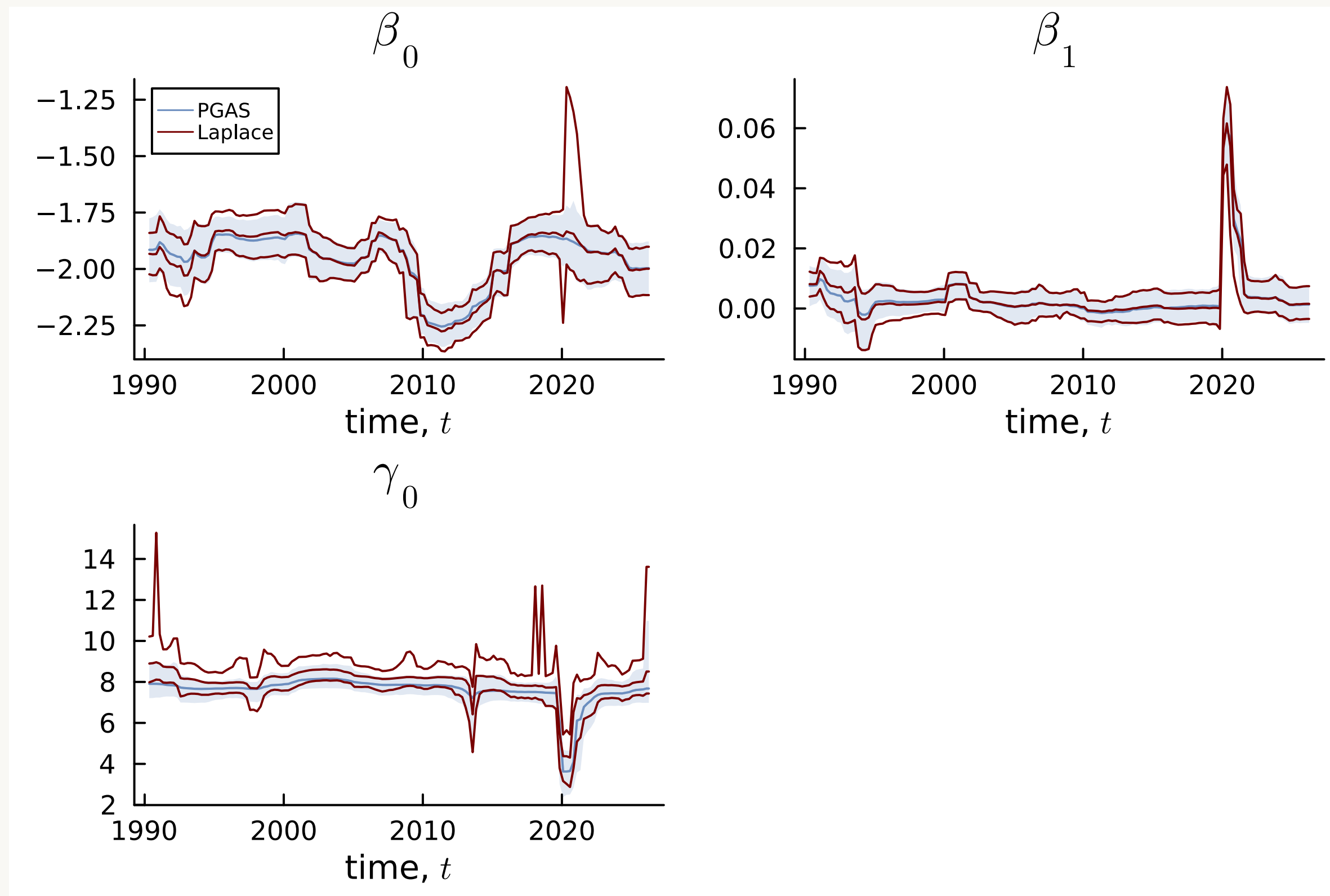
where $\tilde{\boldsymbol{\theta}}_{t|t}$ is posterior mode and $\tilde{\boldsymbol{\Sigma}}_{t|t}$ is inverse of observed information.
 - **Automatic differentiation** makes it automatic.

Time-varying Beta regression for US layoff proportion

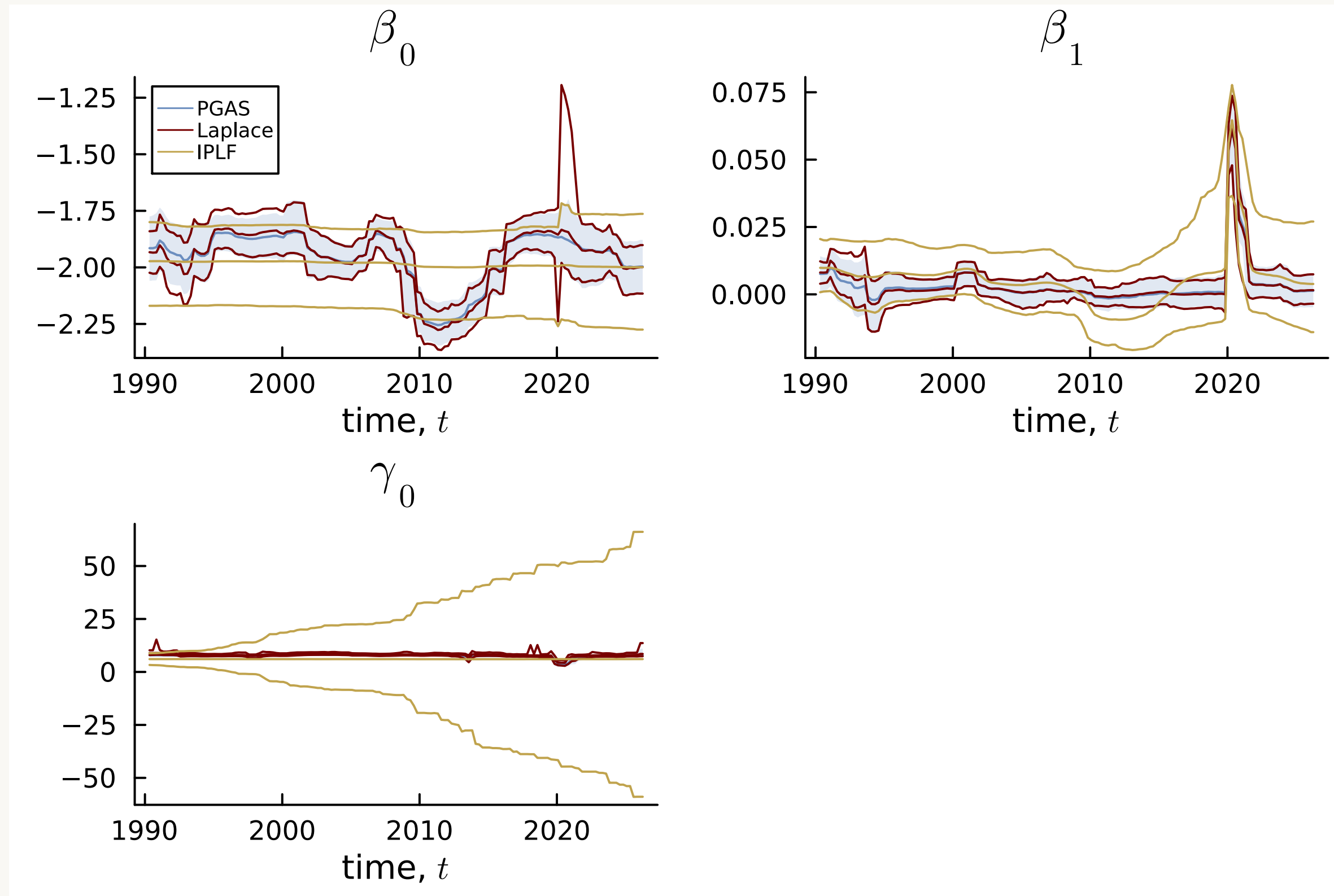
- Monthly US data from 1990:1-2026:4 on:
 - **layoff** - job losers on layoff as % of total unemployed
 - **vix** - CBOE Volatility Index (“fear index”)



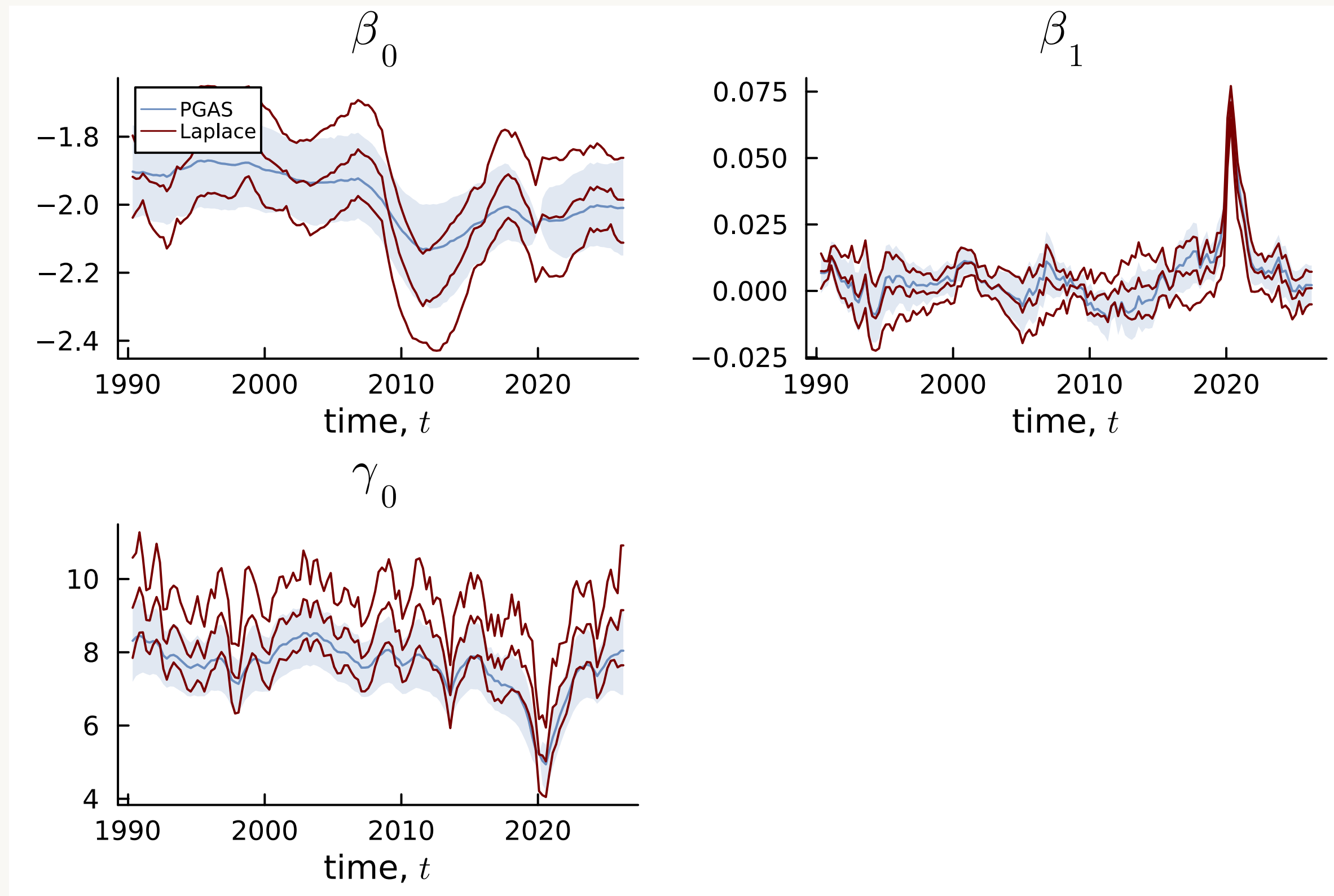
Beta regression for US layoff proportion - dsp prior



Linear approximations struggle with Beta regression

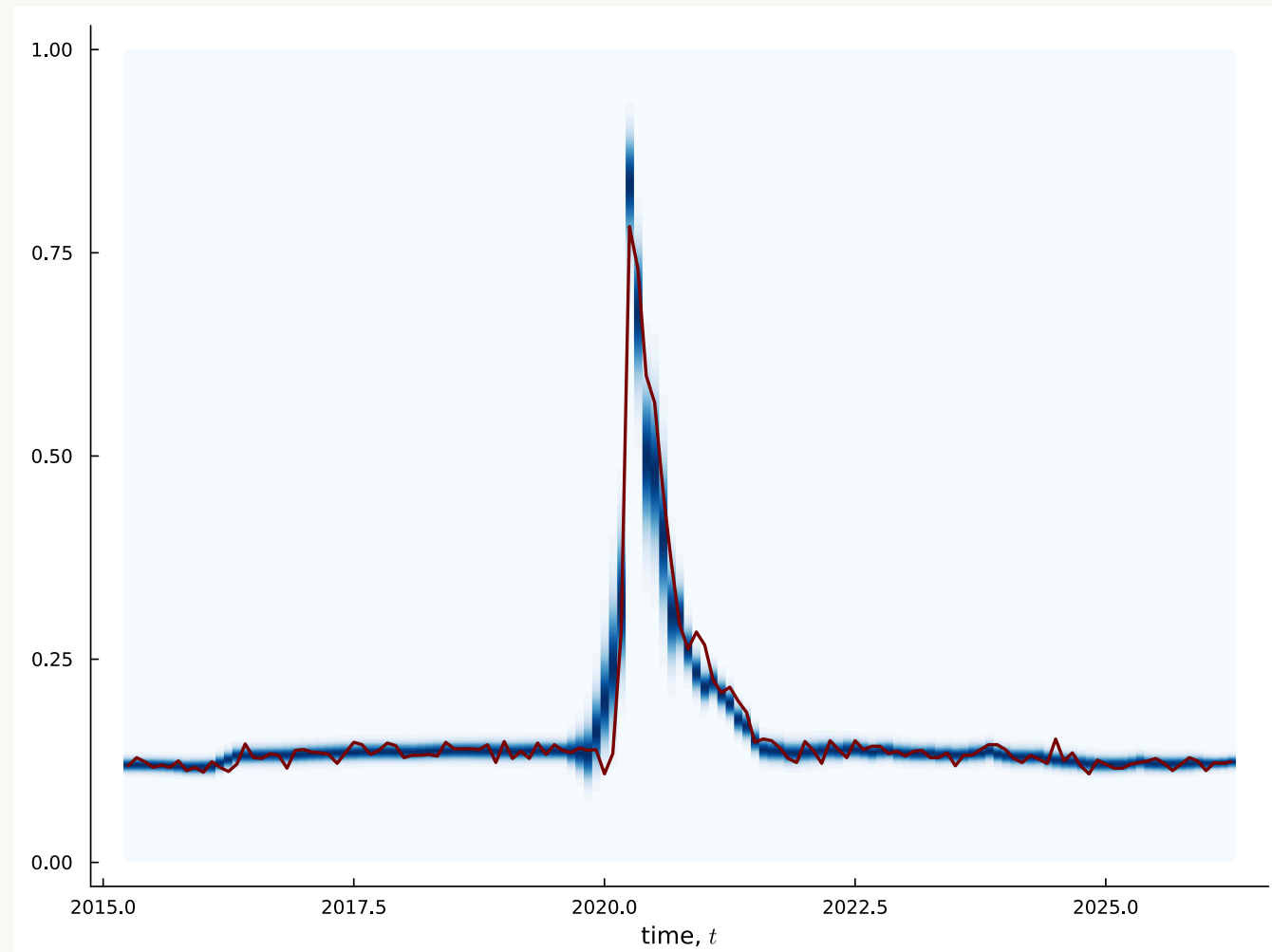


Beta regression for US layoff proportion - homoscedastic Gauss prior

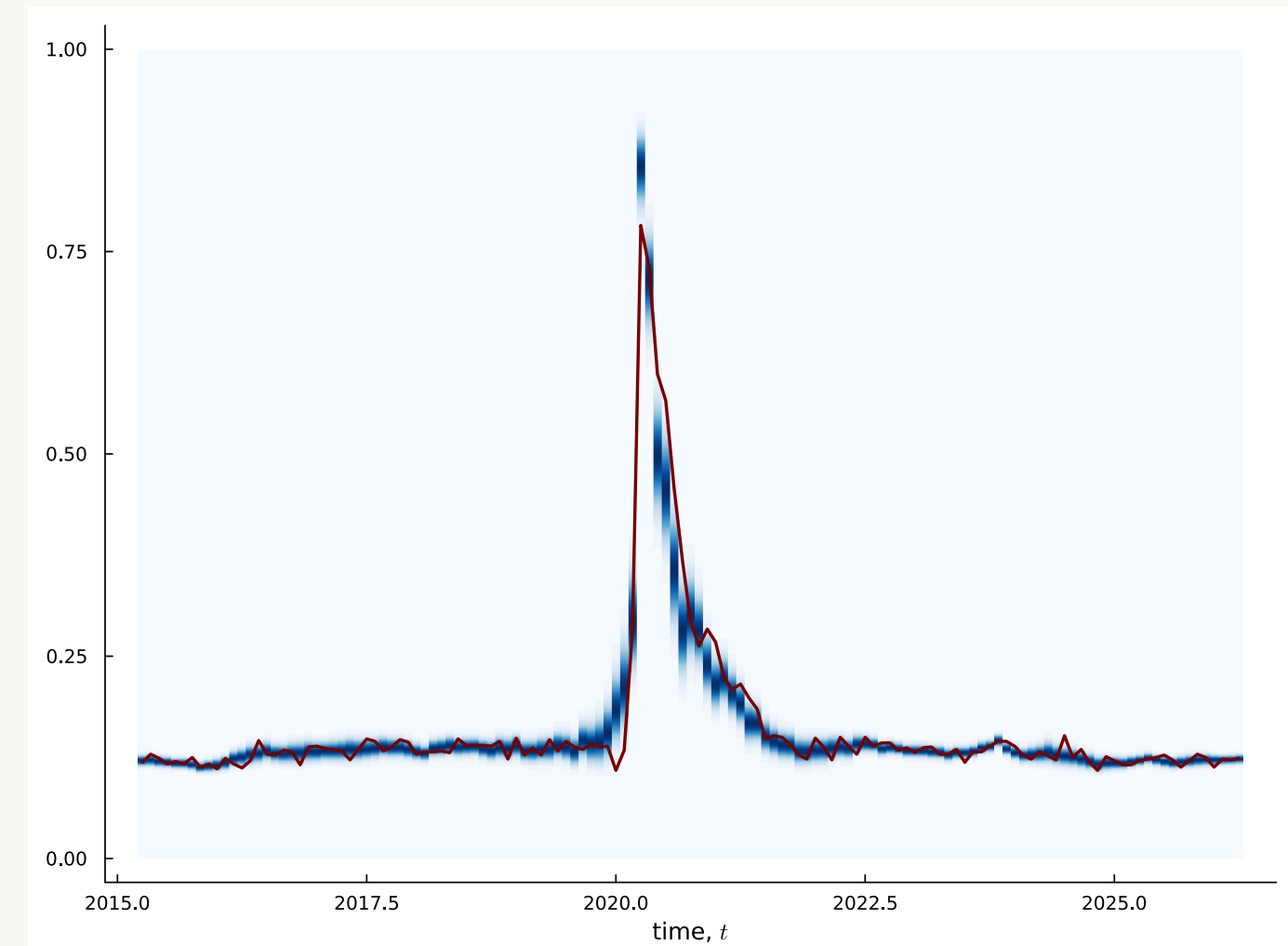


Beta regression for US layoff - fit at posterior median around pandemic

DSP



Homoscedastic Gaussian



Grouping observations for speed and Gaussianity

- Gaussian approximations at every time t .
- How Gaussian are the **likelihood contributions** $p(y_t|\boldsymbol{\theta}_t)$ in $\boldsymbol{\theta}_t$?
- No asymptotic Bernstein-von Mises effect to help.
- A tight propagated prior $\boldsymbol{\theta}_t \sim N(\hat{\boldsymbol{\theta}}_{t|t-1}, \boldsymbol{\Omega}_{t|t-1})$ helps a lot.
- But DSP can give large innovations, so $\boldsymbol{\Omega}_{t|t-1}$ can be large.
- **Group observations** in G groups of J_g observations in group g .
- **Same state $\boldsymbol{\theta}_g$ for observations within a group.**

“When all the hope is gone, there is no reason for pessimism”.

Aki Kaurismäki

$$y_{gj}|\mathbf{x}_{gj} \sim \text{ExpFamily}\left(k^{-1}(\mathbf{x}_{gj}^\top \boldsymbol{\beta}_g), \varphi_g\right)$$

$$\boldsymbol{\beta}_g = \boldsymbol{\beta}_{g-1} + \boldsymbol{\nu}_g, \quad \boldsymbol{\nu}_g \stackrel{\text{iid}}{\sim} N(\mathbf{0}, \text{Diag}(\exp(\boldsymbol{\tau}^2 \odot \mathbf{h}_g)))$$

$$\mathbf{h}_g = \Phi \mathbf{h}_{g-1} + \boldsymbol{\eta}_g, \quad \boldsymbol{\eta}_g \stackrel{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1)$$

where $j = 1, \dots, J_g$ for $g = 1, \dots, G$.

Grouping observations

- Related ideas:
 - Improve Gaussianity in Subsampling MCMC/HMC [18]
 - Improve linearity/Gaussianity in filtering [19]
 - Modeling moderate time evolutions [20]
- Two views:
 - interpolation for algorithmic performance
 - genuine model change: different time scales for observations and parameters.
- **Parameter evolution is automatically adapted** to different group sizes.
- **Massive speed-up**, particularly for sub-daily data.
- Example: 30 min electricity price. Group $48 \times 30 = 1440$ observations by month.
- Large groups: Sequential filtering of individual observations within group.

Natural parameter innovations

- Parameter evolution should adapt to:
 - Scales of covariates
 - Link functions
 - Model geometry
- **Natural gradients** [21, 22] in optimization. Fisher metric.
- Parameter evolution with **natural innovations**

$$y_t | \mathbf{x}_t \sim \text{ExpFamily} \left(k^{-1}(\mathbf{x}_t^\top \boldsymbol{\beta}_t), \varphi_t \right)$$

$$\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} + \mathbf{S}(\boldsymbol{\beta}_{t-1}) \boldsymbol{\nu}_t, \quad \boldsymbol{\nu}_t \stackrel{\text{iid}}{\sim} N(\mathbf{0}, \text{Diag}(\exp(\boldsymbol{\tau}^2 \odot \mathbf{h}_t)))$$

$$\mathbf{h}_t = \Phi \mathbf{h}_{t-1} + \boldsymbol{\eta}_t, \quad \eta_{tk} \stackrel{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1)$$

where $\mathbf{S}(\boldsymbol{\beta}_{t-1}) = \bar{I}^{-1/2}(\boldsymbol{\beta}_{t-1})$

- $\bar{I}(\boldsymbol{\beta}) = \frac{1}{T} I(\boldsymbol{\beta})$ is the **Fisher information** from an “average” observation.
- Non-linear transition model. Prior propagation step: $\mathbf{S}(\boldsymbol{\beta}_{t-1}) \approx \mathbf{S}(\hat{\boldsymbol{\beta}}_{t-1|t-1})$.

Ongoing and Future work

- **Multivariate** extensions.
- **Forecasting**.
- More **efficient** sampling of dsp.
- Better **prior elicitation** for dsp.
- **Metropolis-Hastings correction** by blocking the state. Filtering version of [16].
- **Delayed rejection**:
 - first try with cheap proposal (e.g. EKF)
 - if rejected try more expensive proposal (e.g. Iterated EKF with line search)

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